# LIBRARY MANAGEMENT SYSTEM

## Entity–Relationship Diagram Project

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## Declaration

We hereby declare that this project titled 'Library Management System – ER Diagram' is an original work carried out by us under the guidance of Mr. Atul Kumar Singh. This project has not been submitted previously for any academic evaluation.

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## 1. Introduction

The Library Management System (LMS) is designed to manage records of books, members, librarians, loans, authors, and fines within a library. The system ensures efficient tracking of book issuance, return processes, and fine generation. The ER diagram provides a conceptual representation of the database structure.

## 2. Entities and Attributes

### LIBRARIAN

- Librarian_ID (Primary Key)
- Name
- First_Name(Derived value from Name)
- Middle_Name(Derived value from Name)
- Last_Name(Derived value from Name)
- Mobile_No(Multivalued-can register more than one no.)

### BOOK

- Book_ID (Primary Key)
- Title
- Publisher
- Librarian_ID (Foreign Key)

### MEMBER

- Member_ID (Primary Key)
- Name
- First_Name(Derived value from Name)
- Middle_Name(Derived value from Name)

- Last_Name(Derived value from Name)
- Email
- Mobile_No(Multivalued-can register more than one no.)
- DOB(Date Of Birth)
- Age (Derived Attribute—can be derived from DOB)

### LOAN

- Loan_ID (Primary Key)
- Member_ID (Foreign Key)
- Book_ID (Foreign Key)
- Issue_Date
- Due_Date
- Loan_Status

### FINE(Weak Entity : One Loan → At most One Fine)

- Loan_ID (Foreign Key)
- Fine_Amount
- Payment_Status

## 3. Relationships

1. MANAGES (LIBRARIAN – BOOK): One librarian manages multiple books (1:M).
2. BORROWS (MEMBER – BOOK): Many members can borrow many books (M:M).
3. HAS (MEMBER – LOAN): One member can have many loans (1:M).

4. GENERATES (LOAN – FINE): A loan may generate a fine (1:1, conditional).

## 4. Relational Schema

LIBRARIAN(Librarian_ID PK, First_Name, Middle_Name, Last_Name, Mobile_No)

BOOK(Book_ID PK, Title, Publisher, Librarian_ID FK)

AUTHOR(Author_ID PK, First_Name, Middle_Name, Last_Name)

BOOK_AUTHOR(Book_ID PK/FK, Author_ID PK/FK)

MEMBER(Member_ID PK, First_Name, Middle_Name, Last_Name, Email, Mobile_No, DOB)

LOAN(Loan_ID PK, Member_ID FK, Book_ID FK, Issue_Date, Due_Date, Loan_Status)

FINE(Fine_ID PK, Loan_ID FK, Fine_Amount, Payment_Status, Generate_Date, Generate_Time)

## 5. SQL Table Creation Queries

```
CREATE TABLE LIBRARIAN (
    Librarian_ID INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Middle_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Mobile_No VARCHAR(15)
);

CREATE TABLE BOOK (
```

```
    Book_ID INT PRIMARY KEY,
    Title VARCHAR(100),
    Publisher VARCHAR(100),
    Librarian_ID INT,
    FOREIGN KEY (Librarian_ID) REFERENCES
LIBRARIAN(Librarian_ID)
);

CREATE TABLE MEMBER (
    Member_ID INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Middle_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Mobile_No VARCHAR(15),
    DOB DATE
);

CREATE TABLE LOAN (
    Loan_ID INT PRIMARY KEY,
    Member_ID INT,
    Book_ID INT,
    Issue_Date DATE,
    Due_Date DATE,
    Loan_Status VARCHAR(20),
    FOREIGN KEY (Member_ID) REFERENCES
MEMBER(Member_ID),
    FOREIGN KEY (Book_ID) REFERENCES BOOK(Book_ID)
);

CREATE TABLE FINE (
    Loan_ID INT,
    Fine_Amount DECIMAL(8,2),
```

```
    Payment_Status VARCHAR(20),

    FOREIGN KEY (Loan_ID) REFERENCES LOAN(Loan_ID)
);
```

## #Output Tables:

```
+-------------+
| table_name |
+-------------+
| librarian  |
| book       |
| member     |
| loan       |
| fine       |
+-------------+
5 rows in set (0.00 sec)
```

Table list

```sql
DESC librarian;
DESC book;
DESC member;
DESC loan;
DESC fine;
```

Showing table of every entity

```
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| Librarian_ID | int         | NO   | PRI | NULL    |       |
| First_Name   | varchar(50) | YES  |     | NULL    |       |
| Middle_Name  | varchar(50) | YES  |     | NULL    |       |
| Last_Name    | varchar(50) | YES  |     | NULL    |       |
| Mobile_No    | varchar(15) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Table of librarian

```
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Book_ID     | int          | NO   | PRI | NULL    |       |
| Title       | varchar(100) | YES  |     | NULL    |       |
| Publisher   | varchar(100) | YES  |     | NULL    |       |
| Librarian_ID | int         | YES  | MUL | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Table of book

```
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Member_ID   | int          | NO   | PRI | NULL    |       |
| First_Name  | varchar(50)  | YES  |     | NULL    |       |
| Middle_Name | varchar(50)  | YES  |     | NULL    |       |
| Last_Name   | varchar(50)  | YES  |     | NULL    |       |
| Email       | varchar(100) | YES  |     | NULL    |       |
| Mobile_No   | varchar(15)  | YES  |     | NULL    |       |
| DOB         | date         | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

Table of member

```
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| Loan_ID     | int         | NO   | PRI | NULL    |       |
| Member_ID   | int         | YES  | MUL | NULL    |       |
| Book_ID     | int         | YES  | MUL | NULL    |       |
| Issue_Date  | date        | YES  |     | NULL    |       |
| Due_Date    | date        | YES  |     | NULL    |       |
| Loan_Status | varchar(20) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

Table of loan

```
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| Loan_ID       | int          | NO   | PRI | NULL    |       |
| Fine_Amount   | decimal(8,2) | YES  |     | NULL    |       |
| Payment_Status| varchar(20)  | YES  |     | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

Table of fine

## 6. Conclusion

The Library Management System ER model effectively represents the database structure required for managing library operations. It minimizes redundancy and maintains data consistency.